

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 50

Code of Conduct

I JASMITHA R(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: JASMITHA R

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

6. Data quality refers to the accuracy, completeness, consistency, validity and reliability of data used for analysis. In a large-scale analytics platform, data is collected from multiple source such as databases, website, mobile applications, IoT devices and cloud system.

Major Data Quality Issues:-

1. Inconsistency

Data inconsistency occurs when the same information is represented differently in different system.

Eg:-

- One system stores gender as Male/Female
- Another stores it as M/F.
- Another stores it as 1/0.

Impact:-

- Incorrect aggregation
- Difficulty in joining datasets
- Incorrect reports
- Reduced analytical accuracy.

2. Duplicate Data:-

Duplicate Data occurs when the same record is stored multiple times.

Eg:- A customer may appear three times in a customer database because of repeated registration.

Impact:

- Inflated customer counts
- Incorrect sales statistics
- Increased storage requirements
- Biased machine-learning models

3. Missing Values:

Missing Values occurs when required data is not available.

Eg: A customer record may contain
Customer ID: 101, Age: —, City: Chennai

Impact:

- Statistical calculations may become inaccurate.
- Machine-learning models may fail
- Important patterns may be missed.

4. Incorrect Data:

Data may contain invalid or incorrect values.

Eg:

- Age = 250
- Salary = -50,000
- Invalid email address.

5. Outdated Data:

Old information may no longer represent the current situation.

Eg: A customer's old address may be used for delivery.

Data Governance strategies:-

1. Data Quality standards:-

Organizations should define standards for:

- Accuracy
- Completeness
- Consistency
- Validity
- Timeliness

2. Data Validation

Validation rules should be applied when data enters the system

For example:-

$\text{Age} > 0$ and $\text{Age} < 120$.

3. Data Deduplication:

- Primary keys
- Matching algorithms
- Record linkage techniques

4. Data cleansing:-

- Removing duplicates
- Correcting errors
- Standardizing formats
- Handling missing values.

5. Master Data Management

A single trusted version of important data should be maintained.

7. A recommendation system suggests movies, TV shows, music or other content, based on user preferences and behavior. Streaming platforms generate huge amounts of data from millions of users. Big data technologies help process this information and generate personalized recommendations at scale.

Architecture of Recommendation System:

1. Data collection.

The system collects:

- Viewing history
- Search history
- Likes and dislikes
- Ratings
- Watch duration
- Skip behaviour

2. Big Data Storage:

Large datasets can be stored using technologies such as:

- Hadoop HDFS
- cloud object storage
- NoSql databases.

3. Distributed processing

Frameworks such as Apache spark can process large dataset across multiple machines.

4. User Behavior Analysis:

The system analyse patterns such as:-

- Frequently watched genres
- Favorite actors
- Preferred language

5. Recommendation Techniques:

Recommendations are generated based on similar users.

6. Machine learning:

It can identify hidden patterns in user behavior.

The system can predict:

Probability (user watches content X)

7. Real time processing:

streaming platforms can process current activity.

How Distributed processing Improves the system:

1. Scalability:

Thousands or millions of users can be handled simultaneously.

2. Faster processing:

Data processing is distributed among multiple computers, reducing processing time.

3. Fault Tolerance:-

If one node fails, other nodes can continue processing the workload.

4. Large-Scale Data Analysis:-

Huge amounts of viewing and interaction data can be processed efficiently.

5. Better personalization:-

More user behavior data allows the system to understand individual preferences.

Impact on User Engagement:-

Accurate recommendations:-

- Increase viewing time
- Reduce content search time
- Improve customer satisfaction
- Increase user retention

8. cloud computing provides computing resources such as storage, processing power, databases and networking through the Internet. Modern big data platforms use cloud infrastructure to store and process huge volumes of data without maintaining large physical data.

Role of cloud computing:-

1. cloud storage:-

- structured data
- semi-structured data
- logs
- Images

2. Cloud-Based processing:
Large datasets can be processed using distributed computing services.

2. Data Analytics:

- Data analysis
- Machine learning
- Business Intelligence

4. Real Time Analytics

- IoT devices
- Applications
- websites

Advantages:

1. Scalability:
cloud resources can be increased or decreased according to workload.
2. Cost Management:
organizations do not necessarily need to purchase.
3. Flexibility:
organizations can choose different
 - storage
 - Databases
4. High Availability
cloud providers usually offer distributed infrastructure and backup mechanisms.

7. Transportation and logistics systems generate large amounts of real-time data from GPS devices, vehicles, traffic systems, sensors, mobile application and weather services. Real-time big data analytics can use this information to optimize routes, reduce fuel consumption.

proposed framework:-

1. Data sources

The framework collects data from:-

- GPS devices.
- Vehicle sensors
- Traffic systems
- Weather services

2. Data Ingestion

streaming technologies can collect continuous data from vehicles.

For example:-

Vehicle GPS → Data stream → Analytics platform.

3. Data storage:-

- Distributed file system
- cloud storage.

4. Real-time processing:-

Technologies such as Apache kafka and Apache spark streaming.

5. Route optimization

The system analyse:-

- Traffic congestion
- Road distance
- Accidents
- weather

6. Fuel efficiency:-

The system can analyse:-

- Fuel consumption
- Vehicle speed
- Engine performance
- Idling time

7. Predictive Maintenance:-

Vehicle sensors data can be analyzed to identify possible failures before they occur.

8. Fleet Management:-

Managers can monitor.

- Vehicle location
- Driver performance
- Delivery status
- Fuel consumption

9. Delivery optimization:-

The system can determine the best order for multiple deliveries based on.

- Distance
- Traffic

10. Real-Time Alerts

The system can generate alerts for:

- Traffic congestion
- Vehicle breakdown
- Delayed delivery
- Excessive fuel consumption.

Benefits of Big Data Technologies:-

Route optimization:-

Real-time traffic and GPS data help identify faster routes.

Fuel Efficiency:-

Shorter and less congested routes reduce fuel consumption.

Reduced Delivery Time

Dynamic routing helps avoid delays.

Better Decision-Making:

Managers receive real-time information for operational decisions.

Reduced operational cost:-

Efficient vehicle usage reduces fuel, maintenance, and labor costs.

Improved customer satisfaction:-

Accurate delivery predictions and reduced delay improves customer experience.

10.

Big data analytics enables organizations to collect and analyse enormous amounts of information about individuals and organizations.

Ethical challenges:-

1. Privacy:-

Organizations may collect sensitive personal information without users fully understanding how it will be used.

2. Lack of Consent:-

Users may not provide meaningful consent before their data is collected or analyzed.

3. Data Misuse

Data collected for one purpose may be used for another purpose without user permission.

4. Lack of Transparency:-

Users may not know:-

- what data is collected
- why it is collected
- who uses it
- How long it stored.

6. Algorithmic Bias:-

Machine-learning algorithms can produce unfair results if training data contains bias.

6. Discrimination:-

Analytics may be used to make decisions that unfairly discriminate against individuals.

7. Security Breaches:-

Poor security can expose personal information to unauthorized users.

8. Re-identification

Even an anonymized dataset may sometimes be combined.

Legal challenges:-

organizations must comply with applicable data protection and privacy laws.

Important legal requirements:-

- Lawful data collection
- User Consent where required
- Purpose limitation
- Data minimization
- Data security.